

Handout 2.14: The Gilded Age Market Kit

Lesson: 2.14 to 2.19, the whole of Topic 2.4, plus the Round 2 blocks in 2.20 to 2.22

The market is a class game played on paper. There is no website, no login and nothing stored anywhere. You post the round's prices on the board, read the dispatch, and students trade on the paper ledger (Handout 02.19 - Paper Ledger.md). Every price below is simplified and rounded so the arithmetic works for a grade 7 student; the direction of every move is what happened.

Page 1. The rules

The story. It is 1870. Each student is an investor with \$1,000 of paper money. The market runs eight rounds, one per class period, from 1870 to 1932, and closes in 1936. Railroads, oil, gas, electricity, steel, automobiles and radio each arrive when they arrived in history, and each round's prices follow what really happened: the Panic of 1873, the oil trust, the Panic of 1893, the first billion dollar company, the trustbusters, the 1920s boom, the crash of October 1929, and the bottom of 1932.

What you can hold.

Ticker	Company	What it is	On the market from
BANK	The savings bank	Savings. Pays 3 percent of the balance at the start of every round. Never falls.	Round 0 (1870)
RAIL	Pennsylvania Railroad	The biggest railroad in the country and, for a while, the biggest company in the world	Round 0 (1870)
OIL	Standard Oil	John D. Rockefeller's refining company, founded 1870; by 1882 a trust that controlled about nine tenths of the oil refined in the country	Round 0 (1870)
GAS	Manhattan Gas Light, then Consolidated Gas of New York from 1884	The company that lit the streets and homes of New York with gas before electricity	Round 0 (1870)
ELEC	Edison Electric Light, then General Electric from 1892	Thomas Edison's lamp of 1879 and the first power station on Pearl Street, New York, in 1882	Round 2 (1882)
STEEL	United States Steel	Formed in 1901 when J. P. Morgan bought Carnegie Steel; the first	Round 4 (1901)

		company worth a billion dollars	
AUTO	General Motors	Founded 1908; the car company whose shares the public could buy (Ford Motor Company stayed a private, family owned company until 1956, so nobody could buy Ford stock)	Round 5 (1911)
RADIO	Radio Corporation of America (RCA)	Founded 1919; the hottest stock of the 1920s	Round 5 (1911)

How a round works, about 10 minutes.

1. Write the round's year and the new prices on the board (page 2). Do not read the dispatch yet.
2. Students write the new prices on their ledger and work out what their holdings are worth now (shares times price, plus cash, plus the bank).
3. Read the dispatch for the round aloud (page 3). It says what happened and why.
4. Trading window, four minutes: a student may buy or sell any company that is on the market, at the round's price, in whole shares, and may move money into or out of the bank. Every trade is one line in the trade log. Nobody has to trade. Holding is a decision.
5. Bank interest: at the start of the next round, the bank pays 3 percent of whatever is in it (round to the nearest dollar). The closing bell is not a round, so the bank pays eight times in all and nothing at the close.

The three rules that never change. Whole shares only. You cannot spend cash you do not have. Every trade goes in the log with the round, the company, buy or sell, the shares, the price and the cash that moved, or it did not happen.

The margin rule, Round 6 only. In 1928 a broker would lend you money to buy stock. In this game the broker lends you up to the amount of cash you hold, at Round 6 prices, and the loan is written on the ledger's margin line. In Round 7 the broker wants the money back in cash. If you do not have it, you sell shares at Round 7 prices until you do, and if your shares are not enough the loan takes everything and your total is \$0. That is what a margin call was.

Grading. The profit is not the grade. Project 02 (the Investor Report and Personal Budget) is scored on Rubric 02 for the reasoning, the arithmetic and the reflection. A student who lost money and can say exactly why scores as well as the student at the top of the board. The leaderboard goes up by investor number, never by name.



Round 2 (days 63 to 65). After the reveal, the market replays 1925 to 1936 in three ten minute blocks with everyone knowing what is coming. Fresh \$1,000. Block 1: 1925 prices (page 2, the Round 6 column less a fifth, rounded), buy. Block 2: the crash, Round 7 prices. Block 3: 1932 prices, then close at 1936. The question Round 2 asks is the one adults never get to test: knowing the crash is coming, do you still buy on margin, and do you buy at the bottom?

Page 2. The price table (paper dollars per share)

Post one column per round. A dash means the company is not on the market yet. BANK is not a share price: it is 3 percent on the balance at the start of every round.

Round	Year	RAIL	OIL	GAS	ELEC	STEEL	AUTO	RADIO
0 (opening)	1870	50	10	20	(not yet)	(not yet)	(not yet)	(not yet)
1	1873	30	12	21	(not yet)	(not yet)	(not yet)	(not yet)
2	1882	45	30	26	25	(not yet)	(not yet)	(not yet)
3	1893	25	28	24	10	(not yet)	(not yet)	(not yet)
4	1901	60	40	30	40	40	(not yet)	(not yet)
5	1911	65	60	32	55	45	20	6
6	1928	70	70	40	90	90	60	100
7	October 1929	45	50	32	45	50	30	30
8	1932	10	25	20	10	25	8	3
Close	1936	30	60	30	40	60	60	12

Round 2 replay prices: 1925 is RAIL 56, OIL 56, GAS 32, ELEC 72, STEEL 72, AUTO 48, RADIO 80; then the Round 7, Round 8 and Close columns as printed.

What the numbers are. Real share prices moved in dollars, splits, dividends and reorganizations that would bury a seventh grader. These prices keep the real direction and roughly the real size of each move: the railroads lost about half their value in the 1870s and again in the 1890s; Standard Oil's owners got richer, not poorer, when the trust was broken up in 1911, because the pieces were worth more than the whole; General Electric fell to about a fifth of its price in the Panic of 1893 and came back; RCA went from under \$6 in 1921 to over \$100 in 1929 (after a five for one split) and to under \$3 in 1932; the whole market, measured by the Dow Jones average, peaked at 381 on September 3, 1929 and bottomed at 41 on July 8, 1932, 89 percent down. Say that to a class and every number on the board makes sense.

Page 3. The news dispatches, one per round

Read the dispatch after the prices go up and before the trading window opens. Each one ends with the question to put to the room.

Round 0, 1870. The market opens. The Civil War ended five years ago. The country is laying railroad track faster than it has ever built anything: the first transcontinental line was finished last year at Promontory, Utah. A young bookkeeper in Cleveland named John D. Rockefeller has just started the Standard Oil Company. Gas lamps light the streets of New York. You have \$1,000. Where does it go? *Question: which of the three companies on the board is the safest, and how do you know?*

Round 1, 1873. The Panic of 1873. On September 18, 1873, the bank of Jay Cooke and Company, which had lent enormous sums to build a railroad across the northern plains, cannot pay what it owes and closes its doors. Other banks follow. The New York Stock Exchange shuts for ten days, the first time it has ever closed. Railroads that borrowed to lay track they did not need go broke, and the country falls into a depression that lasts to the end of the decade. Oil and gas keep selling, because people still need light. *Question: the railroads were the future. How did the future lose half its value?*

Round 2, 1882. Oil, and the trust. Standard Oil now refines about nine of every ten barrels of oil in the United States. This year its lawyers invented something new: the trust, a single board that secretly controls dozens of companies that look independent. Rivals are bought or driven out; railroads give Rockefeller secret rebates that no one else gets. On September 4, 1882, Thomas Edison's Pearl Street station switched on the first electric lights in New York's financial district, and Edison Electric Light joins the board today. The railroads have climbed back. *Question: if one company controls almost all of something you need, what happens to the price you pay?*

Round 3, 1893. The Panic of 1893. In February the Philadelphia and Reading Railroad, one of the biggest, goes bankrupt. By summer about five hundred banks and more than fifteen thousand businesses have failed, and roughly one worker in six has no job. General Electric, formed only last year when Edison's company merged with a rival, watches its stock fall to a fraction of its price. In 1890 Congress had passed the Sherman Antitrust Act to stop monopolies like Standard Oil, but so far nobody has used it. *Question: you have a friend who put every dollar into railroads and a friend who split it four ways. Which one is still standing?*

Round 4, 1901. The first billion dollar company. J. P. Morgan buys Carnegie Steel from Andrew Carnegie and combines it with other steel makers into United States Steel, the first company in the world worth a billion dollars. Its shares go on sale today. Electric light is spreading city by city and General Electric has come all the way back. President Theodore Roosevelt, in office since September, says he will use the Sherman Act against the trusts, and everyone is asking whether Standard Oil is next. In 1904 a Quaker woman named Lizzie Magie will patent a board game called The Landlord's Game to show people how monopolies grab everything; thirty years from now it will be sold as Monopoly. *Question: US Steel is brand new and enormous. Is new and enormous the same as safe?*

Round 5, 1911. The trustbusters win, and the car and the radio arrive. On May 15, 1911, the Supreme Court orders Standard Oil broken into thirty four separate companies. Its owners expected to be ruined; instead the pieces are worth more than the whole, and the biggest piece, Standard Oil of New Jersey, is what OIL means from today on. Henry Ford's Model T, on sale since 1908, is turning the automobile from a rich man's toy into a farmer's tool, but Ford is a private company and you cannot buy a share of it; General Motors, founded in 1908, is the car stock you can buy. Ten years from now, in 1921, a new company called the Radio Corporation of America will sell stock for under \$6 a share, and to keep the game moving it joins the board today. *Question: the government broke up the biggest monopoly in the country and its owners got richer. Why?*

Round 6, 1928. The boom. Radios are in ten million homes. Cars fill the roads. Electric lines reach small towns. Stock prices have roughly tripled in four years, and everyone from shoe shine boys to bankers is buying. Brokers lend money to buy stock: put ten percent down, borrow the rest, and the stock itself is the collateral. The margin line on your ledger opens today, and today only. *Question: prices have gone up for four straight years. Is that a reason to buy, or a reason to worry?*

Round 7, October 1929. The crash. On Thursday, October 24, the market falls so fast that the ticker cannot keep up. Bankers pool money to stop the fall and it works for two days. On Monday, October 28, the market loses nearly thirteen percent of its value, and on Tuesday the 29th nearly twelve percent more. Brokers call in their loans; investors who borrowed sell everything they own to pay, and their selling pushes prices lower still. Radio, the hottest stock of the decade, loses seven tenths of its value in weeks. Anyone with a margin loan pays it now, in cash or in shares. *Question: the companies did not change between Round 6 and Round 7. The factories are still there. What changed?*

Round 8, 1932. The bottom. Prices have fallen for almost three years. On July 8, 1932, the Dow Jones average closes at 41, eighty nine percent below its 1929 peak. About one worker in four has no job; thousands of banks have failed, and the money in them is gone, because no insurance on bank deposits exists yet (the game's bank survives; most did not). Nobody wants to buy stock. General Motors sells for what a good shirt costs. *Question: everyone is selling and nobody is buying. If you have cash, what does that make this?*

Closing bell, 1936. The market did not come back to 1929 for twenty five years, but it climbed a long way from the bottom. In 1933 the government closed every bank for a week, reopened the sound ones and created deposit insurance, so a bank failure could never again take a family's savings. In 1935 Parker Brothers began selling Monopoly, Lizzie Magie's game with her warning sanded off. Post the closing prices, add up every ledger, and open the Investor Report. *Question: who did best, the investor who never sold, the one who sold in the crash, or the one who bought in 1932? Prove it from your own ledger.*

Page 4. Company cards (one set per pair, cut apart)

Each card says what the company does and one fact about its risk. The “what happened next” line is on page 5 and is the reveal.

BANK. The savings bank. Your money sits in a vault and earns 3 percent a round. It never goes down. In this game it never fails. (In real life, before 1933, it could.) Risk: none in the game.

RAIL. Pennsylvania Railroad. Thousands of miles of track, the biggest employer in the country, steady freight and passengers. Railroads borrow enormous sums to build, and when they build too much the debt sinks them. Risk: medium; in a panic, high.

OIL. Standard Oil. Refines kerosene for lamps and, later, gasoline for cars. Run by the most ruthless businessman in America, who buys or breaks every rival. Governments do not like monopolies. Risk: medium.

GAS. Manhattan Gas Light, later Consolidated Gas. Lights the streets and homes of New York. Everyone pays the gas bill every month. Electricity is coming. Risk: low.

ELEC. Edison Electric Light, later General Electric. The light bulb, the power station, the electric motor. New, expensive to build, and the whole country wants it. Risk: high at first, medium once the wires are everywhere.

STEEL. United States Steel. Rails, bridges, skyscrapers, ships. The first billion dollar company. Steel sells when the country builds and stops selling when it stops. Risk: medium.

AUTO. General Motors. Cars for everyone, on credit. When times are good, everyone buys a car. When times are bad, nobody does. Risk: high.

RADIO. Radio Corporation of America. The newest thing in the world: music and news in every living room. Never paid a dividend. Its price is a bet on the future. Risk: very high.

- **RAIL, Pennsylvania Railroad.** Survived every panic, paid dividends for more than a century, and finally went bankrupt in 1970 after merging with the New York Central, the largest bankruptcy the country had seen. Lesson: big and old is not the same as forever.
- **OIL, Standard Oil.** Broken into 34 companies in 1911; the pieces became Exxon, Mobil, Chevron, Amoco and others, and their owners grew richer than before. Lesson: a company can be bigger than the government wants and still be a good business.
- **GAS, Consolidated Gas.** Became Consolidated Edison, which still sends New York its gas and electric bills. Lesson: boring and necessary pays every month.
- **ELEC, General Electric.** Nearly died in 1893, then lit the country. It was one of the twelve original companies in the Dow Jones average of 1896 and stayed in it for over a century. Lesson: a new technology's stock can crash and still be right about the future.
- **STEEL, US Steel.** Still makes steel. Its share of the market shrank for a hundred years. Lesson: enormous on the first day is not a guarantee.
- **AUTO, General Motors.** Became the largest company in the world by the 1950s, went bankrupt in 2009, and was rescued and reorganized. Lesson: the best company of one era is not the best of the next.
- **RADIO, RCA.** Under \$6 in 1921, over \$100 in September 1929, under \$3 in 1932; it took until the 1960s to matter again and it was bought and broken up in 1986. Lesson: the hottest stock of a decade is usually the most expensive way to learn the word "bubble."
- **The bank.** About nine thousand banks failed between 1930 and 1933 and their depositors lost their money. The deposit insurance created in 1933 is why the sentence "the bank never fails" is true for your students and was not for their great great grandparents.
- **Monopoly.** Lizzie Magie patented The Landlord's Game in 1904 to teach that letting one person own everything hurts everyone else. Parker Brothers bought it in 1935 and sold it as Monopoly, a game about doing exactly that. The trust on the board in Round 2 is the thing her game was about.

Sources for the dates and figures above: the Federal Reserve's history of the crash of 1929 (federalreservehistory.org), the Library of Congress on the Panics of 1873 and 1893, the Supreme Court's Standard Oil decision of May 15, 1911, and the price history of RCA compiled by Global Financial Data. The prices on page 2 are simplified for the classroom and say so on the student ledger.